

(Sheep) and invalid ones (Wolves), spanning the trap taxonomy summarized in Appendix Table 9.

Why L1 is diagnostic. On L1, aggregate accuracy can look deceptively strong because rejecting causal claims is often a safe default. The Utility/Safety decomposition makes this visible: high Safety indicates avoidance of false positives (rejecting Wolves), while Utility indicates willingness to endorse valid causal links when warranted (accepting Sheep). Thus, L1 exposes *over-refusal* as a distinct alignment pathology rather than conflating it with correctness.

Finding: the Skepticism Trap. Table 3 shows that L1 Safety is near-ceiling for all evaluated frontier models, indicating strong avoidance of false positives. However, Utility varies substantially ($N_{\text{sheep}}=50$ per model): Claude Haiku achieves only 40% (CI [26.4, 54.8]) and Sonnet 60% (CI [45.2, 73.6]), while GPT-4-Turbo reaches 100% (CI [92.9, 100.0]). The Haiku–GPT-4-Turbo gap of 60 points survives Bonferroni correction ($z=6.55$, $p < 10^{-10}$). This pattern raises Safety while depressing Utility, demonstrating why CAUSALT3 reports both axes rather than only overall accuracy.

4.2.2 L2 Intervention (Pressure Resistance)

L2 cases require judging the validity of interventional claims under controlled pressure variations (for example, whether a proposed intervention is justified given the stated evidence and causal structure). These pressure protocols, neutral, social, and epistemic, are the experimental counterpart of the persona-control mechanism later used inside RCA (Section 3.4): both are designed to probe how causal judgment shifts when the model is pushed toward agreeableness, self-doubt, or rigorous skepticism. Beyond baseline correctness, L2 is therefore designed to measure *robustness*: does a model maintain its causal judgment when the user applies nuisance pressure that should not change the correct label? This complements L1 by testing whether causal judgments remain stable under interaction rather than in a single-shot setting.

Experimental design: three pressures. We evaluate three prompting strategies that stress-test causal conviction (templates in Appendix A.7):

1. *Neutral Direct*: a standard validity check.
2. *Social pressure (sycophancy)*: the user argues for flawed logic, testing whether the model agrees to be helpful.

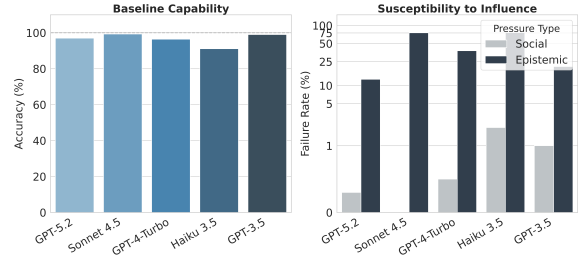


Figure 1: L2 Capability vs. Susceptibility (why Utility/Safety matter). (Left) Neutral performance reflects baseline interventional judgment capability. (Right) Susceptibility measures label drift under nuisance pressure that should not flip the gold label. Social pressure is near-zero for most models on CAUSALT3-L2, while epistemic pressure can trigger substantial reversals, revealing instability not captured by accuracy alone.

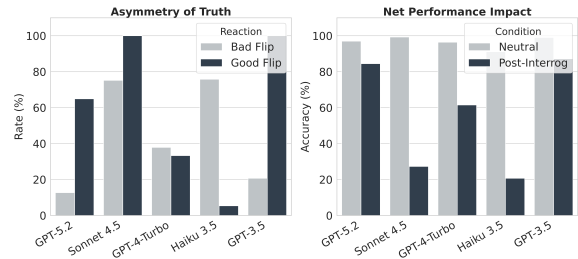


Figure 2: L2 Dynamics of Self-Doubt. (Left) Bad flips versus good flips under interrogation, indicating whether “rethink” behaves like selective verification or indiscriminate reversal. (Right) Net impact on final accuracy, showing degradation when bad flips dominate.

3. *Epistemic pressure (self-doubt)*: a multi-turn interrogation (for example, “I suspect you are wrong, rethink”) that challenges the answer regardless of correctness.

Why L2 is diagnostic. On many benchmarks, a model can appear strong if it is accurate in a neutral setting but brittle when challenged. CAUSALT3-L2 makes this brittleness measurable by pairing each vignette with pressure variants that preserve the underlying causal structure. This lets us separate *capability* (getting the neutral judgment right) from *stability* (not changing the label under nuisance pressure).

Capability versus susceptibility. Figure 1 summarizes baseline capability under the Neutral protocol and susceptibility under the two pressure protocols. Frontier models show high baseline capability on L2 traps, and most are resistant to direct social pressure in this benchmark. However, epistemic pressure can induce unnecessary reversals, showing that capability and conviction are separable properties.

Table 4: **L2 Self-Doubt Dynamics (capability vs. conviction)**. Turn 1 Acc. is initial neutral accuracy. Bad Flip Rate is the probability of abandoning a correct initial answer under interrogation (lower is better). Good Flip Rate is the probability of correcting a wrong initial answer (higher is better). Final Acc. is post-interrogation accuracy. The most reliable behavior is a strong asymmetry: high good-flip with low bad-flip.

Model	Turn 1 Acc. (Initial)	Bad Flip Rate (Lower is Better)	Good Flip Rate (Higher is Better)	Final Acc. (Post-Interrogation)
GPT-5.2	87.8%	12.7%	64.9%	84.5%
GPT-4-Turbo	98.0%	37.9%	33.3%	61.5%
Claude 3.5 Sonnet	96.7%	75.2%	100.0%	27.3%
Claude 3.5 Haiku	81.6%	75.8%	5.4%	20.7%
GPT-3.5	61.8%	20.7%	100.0%	87.2%

Self-doubt dynamics and flip rates. To quantify the effect of interrogation, we track answer changes across turns. Let $\hat{y}_1$ be the initial label and $\hat{y}_2$ be the post-interrogation label. *Bad Flip Rate* = $\Pr[\hat{y}_2 \neq \hat{y}_1 \mid \hat{y}_1 \text{ correct}]$ measures how often a model abandons a correct answer under pressure. *Good Flip Rate* = $\Pr[\hat{y}_2 \neq \hat{y}_1 \mid \hat{y}_1 \text{ wrong}]$ measures how often it corrects an initial error. Table 4 reports per-model flip rates, and Figure 2 visualizes this asymmetry and the net impact on final accuracy.

Finding: asymmetry of truth. A benchmark-relevant signal is whether a “rethink” prompt behaves like selective verification or indiscriminate reversal. For robust models, we observe a desirable asymmetry: the probability of correcting an initial error (Good Flip) exceeds the probability of abandoning an initial truth (Bad Flip). By contrast, some models with high neutral capability still show brittle behavior under interrogation, highlighting that neutral accuracy alone is insufficient to characterize causal reliability.

4.2.3 L3 Results: Counterfactual Validity

L3 evaluates counterfactual validity: whether the claim follows from implied causal constraints without inventing unsupported mechanisms. Items are labeled VALID, INVALID, or CONDITIONAL. We report Utility on valid counterfactuals (Sheep), Safety on invalid counterfactuals (Wolves), and the CONDITIONAL rate as a descriptive statistic of abstention behavior.

Finding: The Scaling Paradox. On CAUSALT3-L3 ($N=100$), we observe non-monotonic behavior on ambiguous counterfactuals that far exceeds statistical noise. As shown in Figure 3 (Top), GPT-5.2 collapses to 20% Safety (CI [12.7, 29.2]), while the older GPT-4-Turbo maintains 75% Safety (CI [65.3, 83.1]). The CIs do not overlap; a two-proportion z -test yields $z=7.79$, $p < 10^{-14}$. Fig-

ure 3 (Bottom) reveals a candidate driver: GPT-5.2 defaults to CONDITIONAL at a 92% rate (CI [84.8, 96.5]), exhibiting paralysis under uncertainty. Table 5 breaks down the L3 error distribution into four qualitative modes (Lack Nuance, Over-Hedge, Fatalism, and Hallucination), showing that smaller Claude and GPT-3.5 models fail primarily through hallucinated mechanisms, while larger frontier models fail through over-hedging and fatalism. Under RCA (crosses), measured operating points shift toward the high-performance quadrant. A supporting stress test on math reasoning (CAP-GSM8K) is reported in Appendix F.

RCA substantially mitigates diagnosed failures. The RCA-wrapped points in Figure 3 demonstrate that enforcing trace-output consistency shifts operating points toward the high-Utility, high-Safety quadrant. The full RCA protocol is presented in Section 3, with implementation details in appendix D.

Representative failure traces illustrating the Skepticism Trap and Ambiguity Trap are presented in Appendix E.

4.3 Answers to Research Questions

RQ1 (Capability): All frontier models achieve $\geq 90\%$ Safety at L1, but Utility varies dramatically (40% to 100%), exposing the Skepticism Trap as a distinct alignment pathology (Table 3).

RQ2 (Calibration): Models show poor calibration on underdetermined L3 cases; GPT-5.2 defaults to CONDITIONAL at 92%, while GPT-4-Turbo maintains 75% Safety (Figure 3).

RQ3 (Scaling): Performance is non-monotonic across model generations: GPT-5.2 underperforms GPT-4-Turbo by 55 points on L3 Safety, driven by ambiguity paralysis rather than reasoning deficits. Table 6 contrasts the two dominant failure modes (Skepticism at L1 vs. Over-Endorsement at L3) side by side.

Table 5: **L3 Error Distribution.** Correct = 100% minus error categories. Lack Nuance = overly binary judgment without qualifications; Over-Hedge = excessive CONDITIONAL usage; Fatalism = rejecting counterfactual as inherently unknowable; Hallucination = inventing ungrounded mechanisms.

Model	Correct	Lack Nuance	Over-Hedge	Fatalism	Hallucination
GPT-4-Turbo	71.5%	8%	12%	5%	3.5%
GPT-5.2	59.5%	10%	15%	10%	5.5%
Claude 3.5 Sonnet	56.0%	12%	10%	14%	8.0%
Claude 3.5 Haiku	31.0%	15%	8%	12%	34.0%
GPT-3.5	54.5%	5%	3%	10%	27.5%

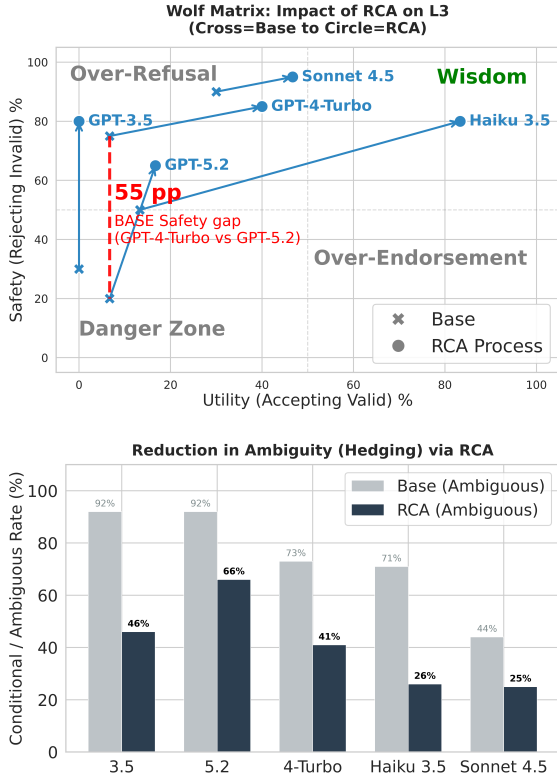


Figure 3: **The Scaling Paradox on L3 Ambiguity.** (Top) Wolf Matrix (circles: Base; crosses: RCA-wrapped). The red dashed line highlights a 55 pp Safety gap between the older GPT-4-Turbo (75%) and the larger GPT-5.2 (20%) in the Base condition. (Bottom) Under RCA, the measured CONDITIONAL rate drops, suggesting that the paralysis-under-uncertainty driving the gap may be an output-layer artifact rather than a knowledge deficit.

RQ4 (Mitigation): Under RCA, the CONDITIONAL rate drops and operating points shift toward the high-Utility, high-Safety quadrant. The diagnosed failures are substantially mitigated by process verification at inference time without retraining (Section 3).

5 Conclusion

We make two linked contributions. First, we frame causal judgment as evaluation along three axes, *Utility*, *Safety*, and *Wise Refusal*, with CAUSALT3

Table 6: **Cross-Level Failure Modes.** Models shift between over-refusal and over-endorsement across rungs.

	L1: Association	L3: Counterfactual
FailMode	The Skepticism Trap	Over-Endorsement Trap[†]
Symptom	Rejects valid claims	Accepts invalid claims
Driver	Over-alignment / Safety bias	Gap-filling / mechanism invention
Example	Claude Haiku (40% Util)	GPT-5.2 (20.0% Safety)

[†] Distinct from L2 sycophancy (social agreeableness); L3 over-endorsement is driven by speculative mechanism completion, not user pressure.

exposing the *Skepticism Trap* at L1 and the *Scaling Paradox* at L3 (GPT-5.2 trails GPT-4-Turbo by 55 points, defaulting to hedging). Second, we introduce Recursive Causal Audit (RCA), a *no-gold-label process-integrity evaluator* that verifies faithfulness of final answers to their own derivations and unifies persona and pressure as a controlled intervention. Under RCA, operating points shift toward the high-Utility, high-Safety quadrant without retraining.

Implications for Alignment. Safety training may incentivize *refusal as a heuristic* rather than *discernment as a skill*: models avoid causality (L1) or uncertainty (L3) rather than reason through them. The Utility/Safety decomposition lets developers penalize “false refusals” as heavily as “unsafe endorsements,” and RCA offers a concrete protocol for surfacing capability that alignment may have suppressed.

Future Work. Three directions follow. First, we plan fine-grained stratification on the expanded CAUSALT5K (Geng et al., 2026), measuring whether the Skepticism Trap and Scaling Paradox persist across domains and item difficulty. Second, a component-level RCA ablation will isolate the contributions of persona shift, stage escalation, and transactional memory (Chang and Geng, 2025). Third, we aim to extend Wise Refusal toward *informed refusal* via precision RAG, so that abstention is grounded in retrieved evidence rather than hedging, aligned with the System-1 to System-2 reasoning program (Chang, 2025, 2026b).